

GATE EE Batch 002

Human Final QA - Mobile Fillable Review Packet

Human reviewer record

Name:

Role or qualification:

Review date (YYYY-MM-DD):

Reviewer identifier (optional):

Type the attestation exactly as printed below:

This packet contains 20 checksum-bound Batch 002 Electric Circuits questions. Revision 2 replaces the rejected ASCII-style formulas with canonical LaTeX and renders them through XeLaTeX.

Canonical source SHA-256: 2a2344b2ed353d0fb309ea98dada882b7371aaa7016e95ff6547d0dba6fb96e5

Strict Formatter report SHA-256: 223878974b5e5999113b8a9a0a87a28d77306cdc2cc1f880108b9cbbfddd4883

Automated prerequisite: **20 PASS / 0 REVIEW / 0 invalid**. This does not replace your human judgment.

Required procedure

1. For each item, solve the question independently before relying on the declared answer.
2. Select exactly one of PASS or FAIL for each of the five checks.
3. Select exactly one final decision: PASS, REVISE, or REJECT.
4. Explain every REVISE or REJECT decision in Notes.
5. Save this filled PDF and return it for conversion into the checksum-bound certification JSON.

Required attestation: I independently reviewed the Batch 002 questions, answers and solutions and approve only the question IDs marked PASS below.

For the qualification field, state your actual background accurately; no diploma or job title is required unless it is genuinely yours.

TMB-GATE-EE-NT-001 - revision 2

Question 1 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Elements

Type / marks / difficulty: NAT / 1 / Easy

Question

A $6\ \mu\text{F}$ capacitor and a $3\ \mu\text{F}$ capacitor are connected in series across an 18 V DC source. Determine the total energy stored by this capacitor network; enter the numerical value in mJ.

Verification reference

Declared answer: 0.324

Independent recomputation evidence: $C_{\text{eq}} = 2\ \mu\text{F}$ and $W = \frac{1}{2}C_{\text{eq}}(18)^2 = 0.324\text{ mJ}$.

Canonical solution:

The series equivalent capacitance is $C_{\text{eq}} = \frac{(6)(3)}{6+3}\ \mu\text{F} = 2\ \mu\text{F}$. The stored energy is $W = \frac{1}{2}C_{\text{eq}}V^2 = \frac{1}{2}(2 \times 10^{-6})(18)^2 = 324 \times 10^{-6}\text{ J} = 0.324\text{ mJ}$. Final answer: 0.324

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-002 - revision 2

Question 2 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Elements

Type / marks / difficulty: MCQ / 1 / Medium

Question

Two coupled inductors in a series-aiding network have $L_1 = 4\text{ H}$, $L_2 = 9\text{ H}$ and coefficient of coupling $k = 0.5$. Determine the equivalent inductance.

Options

- A. 7 H
- B. 13 H
- C. 19 H
- D. 25 H

Verification reference

Declared answer: C

Independent recomputation evidence: $M = 0.5\sqrt{(4)(9)} = 3\text{ H}$ and $L_{\text{eq}} = 4 + 9 + 2M = 19\text{ H}$, option C.

Canonical solution:

The mutual inductance is $M = k\sqrt{L_1 L_2} = 0.5\sqrt{(4)(9)} = 3\text{ H}$. For a series-aiding connection, $L_{\text{eq}} = L_1 + L_2 + 2M = 4 + 9 + 6 = 19\text{ H}$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-003 - revision 2

Question 3 of 20

Route: Electrical Engineering -> Electric Circuits -> Circuit Laws and Analysis

Type / marks / difficulty: MCQ / 2 / Hard

Question

At the input port of a linear network, a $4\ \Omega$ resistor is connected from the port to ground. A dependent current source of value $0.5i_R$, where i_R is the resistor current, is also directed from the port to ground. Determine the resistance seen at the port.

Options

- A. $2\ \Omega$
- B. $\frac{8}{3}\ \Omega$
- C. $4\ \Omega$
- D. $6\ \Omega$

Verification reference

Declared answer: B

Independent recomputation evidence: A test voltage v draws $\frac{v}{4} + 0.5\left(\frac{v}{4}\right) = \frac{3v}{8}$; therefore $\frac{v}{i} = \frac{8}{3}\ \Omega$, option B.

Canonical solution:

Apply a test voltage v at the port. The resistor current is $i_R = \frac{v}{4}$, and the dependent-source current is $0.5i_R$. Hence $i_{\text{in}} = \frac{v}{4} + 0.5\left(\frac{v}{4}\right) = \frac{3v}{8}$, so $R_{\text{in}} = \frac{v}{i_{\text{in}}} = \frac{8}{3}\ \Omega$. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-004 - revision 2

Question 4 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Elements

Type / marks / difficulty: MSQ / 1 / Easy

Question

Select every correct statement about ideal passive network elements under finite, impulse-free excitation; one or more options may be correct.

Options

- A. The energy stored in a capacitor is $\frac{1}{2}Cv^2$.
- B. The energy stored in an inductor is Li^2 .
- C. A capacitor voltage cannot change instantaneously.
- D. An inductor current cannot change instantaneously.

Verification reference

Declared answer: A, C, D

Independent recomputation evidence: The ideal stored energies are $\frac{1}{2}Cv^2$ and $\frac{1}{2}Li^2$; capacitor voltage and inductor current cannot jump under finite, impulse-free excitation.

Canonical solution:

Capacitor energy is $\frac{1}{2}Cv^2$, whereas inductor energy is $\frac{1}{2}Li^2$; therefore option B is missing the factor $\frac{1}{2}$. With finite, impulse-free excitation, capacitor voltage and inductor current cannot change instantaneously. Therefore A, C and D are correct. Final answer: A, C, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-005 - revision 2

Question 5 of 20

Route: Electrical Engineering -> Electric Circuits -> Circuit Laws and Analysis

Type / marks / difficulty: NAT / 2 / Medium

Question

In a resistive network, node v_1 is connected to a 12 V source through $3\ \Omega$, to ground through $6\ \Omega$, and to node v_2 through $2\ \Omega$. Node v_2 is connected to ground through $4\ \Omega$. Using KCL and node-voltage analysis, determine v_2 ; enter the numerical value in V.

Verification reference

Declared answer: 4

Independent recomputation evidence: KCL gives $v_1 = 1.5v_2$ and $6v_1 - 3v_2 = 24$; hence $v_2 = 4\text{ V}$.

Canonical solution:

KCL at v_2 gives $\frac{v_2 - v_1}{2} + \frac{v_2}{4} = 0$, hence $v_1 = 1.5v_2$. KCL at v_1 gives $\frac{v_1 - 12}{3} + \frac{v_1}{6} + \frac{v_1 - v_2}{2} = 0$. Multiplying by 6 yields $6v_1 - 3v_2 = 24$. Substitution gives $6(1.5v_2) - 3v_2 = 24$, so $v_2 = 4\text{ V}$. Final answer: 4

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-006 - revision 2

Question 6 of 20

Route: Electrical Engineering -> Electric Circuits -> Circuit Laws and Analysis

Type / marks / difficulty: NAT / 2 / Hard

Question

Two non-reference node voltages satisfy $v_1 - v_2 = 5\text{ V}$ because of an ideal voltage source between them. A $2\ \Omega$ branch connects v_1 to ground, a $3\ \Omega$ branch connects v_2 to ground, and a 4 A current source injects current into v_2 . Using supernode KCL, determine v_1 ; enter the numerical value in V.

Verification reference

Declared answer: 6.8

Independent recomputation evidence: $\frac{v_1}{2} + \frac{v_2}{3} = 4$ with $v_1 - v_2 = 5$ gives $v_2 = 1.8\text{ V}$ and $v_1 = 6.8\text{ V}$.

Canonical solution:

KCL for the supernode is $\frac{v_1}{2} + \frac{v_2}{3} = 4$. The voltage-source constraint is $v_1 = v_2 + 5$. Substituting and multiplying by 6 gives $3(v_2 + 5) + 2v_2 = 24$, so $v_2 = 1.8\text{ V}$ and $v_1 = 6.8\text{ V}$. Final answer: 6.8

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-007 - revision 2

Question 7 of 20

Route: Electrical Engineering -> Electric Circuits -> Circuit Laws and Analysis

Type / marks / difficulty: MCQ / 2 / Medium

Question

Clockwise mesh currents I_1 and I_2 in a resistor network satisfy $5I_1 - 2I_2 = 8$ and $-2I_1 + 4I_2 = 2$. Determine the current in their common branch, directed with I_1 .

Options

- A. 0.625 A
- B. 1.625 A
- C. 2.250 A
- D. 3.875 A

Verification reference

Declared answer: A

Independent recomputation evidence: The two equations give $I_1 = 2.25$ A and $I_2 = 1.625$ A; hence $I_1 - I_2 = 0.625$ A, option A.

Canonical solution:

Solving the mesh-current equations gives $I_1 = \frac{36}{16} = 2.25$ A and $I_2 = \frac{26}{16} = 1.625$ A. The common-branch current in the I_1 direction is $I_1 - I_2 = 0.625$ A. Final answer: A

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-008 - revision 2

Question 8 of 20

Route: Electrical Engineering -> Electric Circuits -> Circuit Laws and Analysis

Type / marks / difficulty: MSQ / 1 / Easy

Question

Select every correct statement about node-voltage analysis of a connected electrical network; one or more options may be correct.

Options

- A. One node is selected as the reference node.
- B. An ideal voltage source between two non-reference nodes can be handled using a supernode.
- C. For n nodes, at most $n - 1$ independent KCL node equations are required.
- D. The method cannot be used when dependent sources are present.

Verification reference

Declared answer: A, B, C

Independent recomputation evidence: A reference node, a supernode for an inter-node ideal voltage source, and at most $n - 1$ independent KCL equations are valid; dependent sources do not prohibit nodal analysis.

Canonical solution:

Node-voltage analysis selects a reference, uses supernodes for ideal voltage sources between unknown nodes, and requires at most $n - 1$ independent KCL equations. Dependent sources are permitted when their controlling relations are included. Thus A, B and C are correct. Final answer: A, B, C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-009 - revision 2

Question 9 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Theorems

Type / marks / difficulty: MCQ / 2 / Medium

Question

An 18 V ideal source feeds a $3\ \Omega$ series resistor and an output node; a $6\ \Omega$ resistor connects that node to ground. A $4\ \Omega$ load is then connected from the output node to ground. Determine the load current using the Thevenin equivalent of the source network.

Options

- A. 1 A
- B. 2 A
- C. 3 A
- D. 4 A

Verification reference

Declared answer: B

Independent recomputation evidence: $V_{th} = 18 \left(\frac{6}{9} \right) = 12\text{ V}$ and $R_{th} = 3 \parallel 6 = 2\ \Omega$; therefore $I_L = \frac{12}{2+4} = 2\text{ A}$, option B.

Canonical solution:

The open-circuit voltage is $V_{th} = 18 \left(\frac{6}{3+6} \right) = 12\text{ V}$. With the independent source deactivated, $R_{th} = 3 \parallel 6 = 2\ \Omega$. Therefore $I_L = \frac{12}{2+4} = 2\text{ A}$. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-010 - revision 2

Question 10 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Theorems

Type / marks / difficulty: MCQ / 1 / Easy

Question

A Norton network consists of a 3 A current source in parallel with $6\ \Omega$. When a $3\ \Omega$ load is connected across the network, determine the load current.

Options

- A. 0.5 A
- B. 1 A
- C. 2 A
- D. 3 A

Verification reference

Declared answer: C

Independent recomputation evidence: Current division gives $I_L = 3 \left(\frac{6}{6+3} \right) = 2\text{ A}$, option C.

Canonical solution:

By current division, the 3 A Norton current splits between $6\ \Omega$ and $3\ \Omega$. The load current is $I_L = 3 \left(\frac{6}{6+3} \right) = 2\text{ A}$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-011 - revision 2

Question 11 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Theorems

Type / marks / difficulty: NAT / 2 / Hard

Question

A sinusoidal network has Thevenin RMS voltage $V_{th} = 10\text{ V}$ and Thevenin impedance $Z_{th} = (3 + j4)\Omega$. A conjugately matched load is connected. Determine the maximum average load power; enter the numerical value in W.

Verification reference

Declared answer: 8.333

Independent recomputation evidence: Conjugate matching gives $P_{\max} = \frac{|V_{th}|^2}{4R_{th}} = \frac{100}{12} = 8.333\text{ W}$.

Canonical solution:

For conjugate matching, $Z_L = Z_{th}^* = (3 - j4)\Omega$. The maximum average load power is $P_{\max} = \frac{|V_{th}|^2}{4R_{th}} = \frac{10^2}{4(3)} = 8.333\text{ W}$. Final answer: 8.333

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-012 - revision 2

Question 12 of 20

Route: Electrical Engineering -> Electric Circuits -> Network Theorems

Type / marks / difficulty: MSQ / 1 / Medium

Question

Select every correct statement about superposition in a linear electrical network; one or more options may be correct.

Options

- A. An independent ideal voltage source is replaced by a short circuit when deactivated.
- B. An independent ideal current source is replaced by an open circuit when deactivated.
- C. Dependent sources remain active.
- D. The total power cannot in general be obtained by adding the powers caused by the individual sources.

Verification reference

Declared answer: A, B, C, D

Independent recomputation evidence: Independent voltage/current sources become shorts/opens, dependent sources remain, and nonlinear power does not superpose.

Canonical solution:

Superposition deactivates only independent sources: ideal voltage sources become shorts and ideal current sources become opens. Dependent sources remain because they are part of the network model. Voltages and currents superpose, but power is nonlinear, so the separate-source powers do not generally add. All four statements are correct. Final answer: A, B, C, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-013 - revision 2

Question 13 of 20

Route: Electrical Engineering -> Electric Circuits -> Transient Analysis

Type / marks / difficulty: NAT / 2 / Medium

Question

In a first-order RC network, the capacitor voltage is initially 2 V and its steady-state value after switching is 10 V. The resistance is 2 k Ω and the capacitance is 100 μ F. Determine the capacitor voltage one time constant after switching; enter the numerical value in V.

Verification reference

Declared answer: 7.057

Independent recomputation evidence: $\tau = RC = 0.2 \text{ s}$ and $v_C(\tau) = 10 + (2 - 10)e^{-1} = 7.057 \text{ V}$.

Canonical solution:

The time constant is $\tau = RC = (2000)(100 \times 10^{-6}) = 0.2 \text{ s}$. At $t = \tau$, $v_C(\tau) = 10 + (2 - 10)e^{-1} = 10 - \frac{8}{e} = 7.057 \text{ V}$. Final answer: 7.057

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-014 - revision 2

Question 14 of 20

Route: Electrical Engineering -> Electric Circuits -> Transient Analysis

Type / marks / difficulty: NAT / 2 / Medium

Question

A 12 V DC source is applied at $t = 0$ to an initially unenergized series RL network with resistance $R = 4 \Omega$ and inductance $L = 2$ H. Determine the inductor current at $t = (0.5 \ln 2)$ s; enter the numerical value in A.

Verification reference

Declared answer: 1.5

Independent recomputation evidence: $I_{\infty} = 3$ A, $\tau = \frac{L}{R} = 0.5$ s and $\frac{t}{\tau} = \ln 2$; hence $i_L = 3 \left(1 - \frac{1}{2}\right) = 1.5$ A.

Canonical solution:

The final current is $I_{\infty} = \frac{12}{4} = 3$ A, and the time constant is $\tau = \frac{L}{R} = \frac{2}{4} = 0.5$ s. Here $\frac{t}{\tau} = \ln 2$, so $i_L(t) = 3 \left(1 - e^{-\ln 2}\right) = 3 \left(1 - \frac{1}{2}\right) = 1.5$ A. Final answer: 1.5

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-015 - revision 2

Question 15 of 20

Route: Electrical Engineering -> Electric Circuits -> Resonance and Two-Port Networks

Type / marks / difficulty: MCQ / 1 / Easy

Question

A series RLC network has inductance $L = 50 \text{ mH}$ and capacitance $C = 20 \mu\text{F}$. Determine its undamped resonant angular frequency.

Options

- A. 100 rad s^{-1}
- B. 500 rad s^{-1}
- C. 1000 rad s^{-1}
- D. 5000 rad s^{-1}

Verification reference

Declared answer: C

Independent recomputation evidence: $LC = 10^{-6}$ and $\omega_0 = \frac{1}{\sqrt{LC}} = 1000 \text{ rad s}^{-1}$, option C.

Canonical solution:

The resonant angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$. Since $LC = (50 \times 10^{-3})(20 \times 10^{-6}) = 10^{-6}$, $\omega_0 = \frac{1}{10^{-3}} = 1000 \text{ rad s}^{-1}$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-016 - revision 2

Question 16 of 20

Route: Electrical Engineering -> Electric Circuits -> Resonance and Two-Port Networks

Type / marks / difficulty: MCQ / 1 / Medium

Question

A series RLC network has resistance $R = 20\ \Omega$ and inductance $L = 0.1\ \text{H}$. Determine its half-power bandwidth in angular frequency.

Options

- A. $100\ \text{rad s}^{-1}$
- B. $200\ \text{rad s}^{-1}$
- C. $500\ \text{rad s}^{-1}$
- D. $2000\ \text{rad s}^{-1}$

Verification reference

Declared answer: B

Independent recomputation evidence: The series-RLC angular bandwidth is $\Delta\omega = \frac{R}{L} = \frac{20}{0.1} = 200\ \text{rad s}^{-1}$, option B.

Canonical solution:

For series RLC resonance, the angular-frequency bandwidth is $\Delta\omega = \frac{R}{L} = \frac{20}{0.1} = 200\ \text{rad s}^{-1}$. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-017 - revision 2

Question 17 of 20

Route: Electrical Engineering -> Electric Circuits -> Sinusoidal Steady State

Type / marks / difficulty: MSQ / 2 / Hard

Question

A single-phase impedance load absorbs $P = 8 \text{ kW}$ at a lagging power factor of 0.8. Select every correct statement about this sinusoidal steady-state network; one or more options may be correct.

Options

- A. The load reactive power is 6 kVAR.
- B. The apparent-power magnitude is 10 kVA.
- C. The load current leads the load voltage.
- D. A 6 kVAR shunt capacitor would correct the source power factor to unity.

Verification reference

Declared answer: A, B, D

Independent recomputation evidence: $\tan(\cos^{-1} 0.8) = 0.75$ gives $Q = 6 \text{ kVAR}$ and $|S| = 10 \text{ kVA}$; the current lags, and 6 kVAR capacitive compensation reaches unity power factor.

Canonical solution:

With $\cos \phi = 0.8$, $\tan \phi = 0.75$. Hence $Q = P \tan \phi = (8)(0.75) = 6 \text{ kVAR}$ and $|S| = \frac{P}{\cos \phi} = \frac{8}{0.8} = 10 \text{ kVA}$. A lagging load current does not lead its voltage. A 6 kVAR capacitive shunt cancels the inductive reactive power, so A, B and D are correct. Final answer: A, B, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-018 - revision 2

Question 18 of 20

Route: Electrical Engineering -> Electric Circuits -> Balanced Three-Phase Circuits and Complex Power

Type / marks / difficulty: NAT / 2 / Medium

Question

A balanced three-phase, star-connected load has per-phase impedance $Z_{\text{ph}} = (8 + j6) \Omega$ and is supplied at a line-to-line RMS voltage $V_L = 400 \text{ V}$. Determine the total real power in kilowatts.

Verification reference

Declared answer: 12.8

Independent recomputation evidence: $|Z| = 10 \Omega$ and $\cos \phi = 0.8$; the three-phase real power evaluates to 12.8 kW.

Canonical solution:

The phase-voltage magnitude is $V_{\text{ph}} = \frac{400}{\sqrt{3}} \text{ V}$, the phase-current magnitude is $I_{\text{ph}} = \frac{400/\sqrt{3}}{10} \text{ A}$, and the load power factor is $\cos \phi = \frac{8}{10} = 0.8$. Thus $P = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} (400) \left(\frac{400}{10\sqrt{3}} \right) (0.8) = 12800 \text{ W} = 12.8 \text{ kW}$. Final answer: 12.8

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-019 - revision 2

Question 19 of 20

Route: Electrical Engineering -> Electric Circuits -> Resonance and Two-Port Networks

Type / marks / difficulty: NAT / 2 / Hard

Question

A two-port network has $z_{11} = 4\Omega$, $z_{12} = z_{21} = 2\Omega$, and $z_{22} = 5\Omega$. Port 2 is terminated by a load $Z_L = 3\Omega$. Determine the input impedance; enter the numerical value in Ω .

Verification reference

Declared answer: 3.5

Independent recomputation evidence: $Z_{\text{in}} = z_{11} - \frac{z_{12}z_{21}}{z_{22}+Z_L} = 4 - \frac{4}{8} = 3.5\Omega$.

Canonical solution:

For a z -parameter two-port terminated by Z_L , the input impedance is $Z_{\text{in}} = z_{11} - \frac{z_{12}z_{21}}{z_{22}+Z_L}$. Therefore $Z_{\text{in}} = 4 - \frac{(2)(2)}{5+3} = 4 - 0.5 = 3.5\Omega$. Final answer: 3.5

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

TMB-GATE-EE-NT-020 - revision 2

Question 20 of 20

Route: Electrical Engineering -> Electric Circuits -> Balanced Three-Phase Circuits and Complex Power

Type / marks / difficulty: MCQ / 2 / Medium

Question

A balanced three-phase, delta-connected load has branch-impedance magnitude $|Z_{\Delta}| = 12 \Omega$ and lagging phase angle 30° . It is supplied at a line-to-line RMS voltage $V_L = 240 \text{ V}$. Determine the approximate total real power.

Options

- A. 7.20 kW
- B. 12.47 kW
- C. 14.40 kW
- D. 24.94 kW

Verification reference

Declared answer: B

Independent recomputation evidence: The delta phase current is $\frac{240}{12} = 20 \text{ A}$; $3(240)(20) \cos 30^\circ = 12.47 \text{ kW}$, option B.

Canonical solution:

For a delta load, $V_{\text{ph}} = V_L = 240 \text{ V}$, so $I_{\text{ph}} = \frac{240}{12} = 20 \text{ A}$. The total real power is $P = 3V_{\text{ph}}I_{\text{ph}} \cos 30^\circ = 3(240)(20) \cos 30^\circ = 12470.8 \text{ W} \approx 12.47 \text{ kW}$. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness		
Answer correctness		
Solution correctness		
Clarity / ambiguity		
Originality-conflict check		

Decision (select exactly one): PASS REVISE REJECT

Notes / independent calculation summary:

Review completion

Complete this page only after reviewing all 20 questions.

Final reviewed-results decision: APPROVE REVIEWED RESULTS

Reviewer typed signature:

Final comments:

Paper eligibility remains blocked until the completed human record is validated and only explicitly approved IDs are promoted.